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Digital signature system

Abstract

The disclosure includes digital signature systems having dynamically validated, multi-level signature seals. A method for leveraging such a system can include receiving a digital signature such as a seal, stamp, or certification. The digital signature can be a digital facsimile of a physical signature, stamp, or seal made on a physical document. Machine-readable code can be attached to the digital signature to direct a user, upon scanning (i.e., via a camera in the physical space) or selecting (i.e., via clicking in the digital space) the code (or seal), to multi-level digital signature data for the digital signature. The multi-level digital signature data includes an artifact that graphically indicates the seal's validation status. A user is directed to the multi-level digital signature data responsive to scanning or selecting the code. The user manually validates or invalidates the digital signature via a user interface configured to display the multi-level digital signature data.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application claims the priority benefit of U.S. Provisional Application No. 63/347,216 filed on May 31, 2022, the disclosure of which is incorporated herein by reference for all purposes.

INTRODUCTION

(1) The subject disclosure relates to credentialing and authentication systems, and particularly to a digital signature system having dynamically validated, multi-level signature seals.

(2) The use of seals for protecting against fraud and tampering can be traced back to antiquity. Wax seals, for example, have been used by monarchs, administrators, bankers, and merchants for hundreds of years to protect official decrees, sensitive messages, ledgers, and contracts. The wax seal can be unique or otherwise attributable to the bearer and the presence of such a seal serves as a validation or attestation that the sealed document was in fact authored or certified by the bearer.

(3) Equivalent techniques are still used today, although the wax seal has been largely replaced by physical seals made via embossing or stamping and by electronic seals made digitally, such as digital certificates. For example, professional engineers and architects can seal a drawing to indicate that the respective version or revision of the drawing is the final, approved version for construction. Notaries commonly stamp a range of documents as proof that a signature was officially witnessed. A website can include a digital certificate for a range of reasons, such as proof that stored user data is secure, to verify website ownership, and to prevent others from impersonating the website.

SUMMARY

(4) Embodiments of the present invention are directed to a digital signature system having dynamically validated seals. A non-limiting example method for leveraging such a system can include receiving a digital signature such as a seal, stamp, or certification. The digital signature can be a digital facsimile of a physical signature, stamp, or seal made on a physical document. Machine-readable code can be attached to or embedded within the digital signature to direct a user, upon scanning the machine-readable code (e.g., in the case of physical code, such as a QR code,

affixed to the document and/or signature) or upon selecting the machine-readable code (e.g., by clicking the image of the seal in a digital format), to multi-level digital signature data for the digital signature. The multi-level digital signature data includes an artifact configured to graphically indicate the validation status of the respective seal. The method can include directing a user to the multi-level digital signature data responsive to the machine-readable code being scanned or selected (i.e., clicked in a digital format). The user can manually validate or invalidate the digital signature via a user interface configured to display the multi-level digital signature data.

(5) Other embodiments of the present invention implement features of the above-described method in computer systems and computer program products.

(6) Additional technical features and benefits are realized through the techniques of the present invention. Embodiments and aspects of the invention are described in detail herein and are considered a part of the claimed subject matter. For a better understanding, refer to the detailed description and to the drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The specifics of the exclusive rights described herein are particularly pointed out and distinctly claimed in the claims at the conclusion of the specification. The foregoing and other features and advantages of the embodiments of the invention are apparent from the following detailed description taken in conjunction with the accompanying drawings in which:

(2) FIG. 1 depicts a document having a dynamically validated, multi-level signature seal according to one or more embodiments;

(3) FIG. 2A depicts a validation portal for the seal when the seal is valid;

(4) FIG. 2B depicts the validation portal for the seal when the seal is invalid;

(5) FIG. 3 illustrates a first page (“1-Basic”) of a multi-level digital signature validation portal reached by clicking or selecting a multi-level signature validation link in accordance with one or more embodiments;

(6) FIG. 4 illustrates a second page (“2-Advanced”) of the multi-level digital signature validation portal in accordance with one or more embodiments;

(7) FIG. 5 illustrates a third page (“4-Contractual Info”) of the multi-level digital signature validation portal in accordance with one or more embodiments;

(8) FIG. 6 illustrates a fourth page (“3-Quality Control”) of the multi-level digital signature validation portal in accordance with one or more embodiments;

(9) FIG. 7 illustrates a user interface for managing dynamically validated, multi-level signature seals and multi-level digital signature validation portals according to one or more embodiments;

(10) FIG. 8 illustrates an exemplary selectable field of the user interface according to one or more embodiments;

(11) FIG. 9A illustrates an exemplary dashboard for a valid seal after selecting a document in the user interface according to one or more embodiments;

(12) FIG. 9B illustrates an exemplary dashboard for invalidating the seal in the user interface according to one or more embodiments;

(13) FIG. 9C illustrates an exemplary dashboard after invalidating the seal in the user interface according to one or more embodiments;

(14) FIG. 10 is a computer system according to one or more embodiments; and

(15) FIG. 11 is a flowchart in accordance with one or more embodiments.

DETAILED DESCRIPTION

(16) The following description is merely exemplary in nature and is not intended to limit the present disclosure, its application or uses. It should be understood that throughout the drawings,

corresponding reference numerals indicate like or corresponding parts and features. As used herein, the term module refers to processing circuitry that may include an application specific integrated circuit (ASIC), an electronic circuit, a processor (shared, dedicated, or group) and memory that executes one or more software or firmware programs, a combinational logic circuit, and/or other suitable components that provide the described functionality.

(17) Modern credentialing and authentication systems rely on the use of physical seals (stamped, embossed, wax, etc.) and digital seals (digital certificates, etc.) to verify and authenticate physical and digital documents and objects across a range of disciplines including notarization services, engineering, and banking. Unfortunately, there are some issues and limitations inherent to the types of physical and electronic seals currently available.

(18) Physical seals are increasingly subjected to ever more sophisticated counterfeiting techniques. Given the ease of replicating a physical seal using modern equipment, strictly physical seals are becoming less trusted (and therefore less useful) for verifying the authorship or authenticity of a sealed document or object.

(19) In the digital world, certificates and encryption keys have, in some ways, taken the place of physical seals to provide security and/or authenticity for digital documents, pages, programs, and other types of digital media. Traditional digital certificates and encryption keys, however, are static, single level signature objects (e.g., certificates serve as a type of snapshot validation for websites) that are not easily portable to the physical world (e.g., for securing printed documents).

(20) One or more embodiments of the present invention address one or more of the above-described shortcomings by providing a digital signature system having dynamically validated, multi-level signature seals. Embodiments of the present invention provide a mechanism to allow a bearer to dynamically validate and invalidate their digital seals at any time and from any location. In some embodiments, a previously valid seal can be invalidated by the bearer and such invalidation will be dynamically passed to all documents and objects containing that respective seal.

(21) Technical solutions described herein facilitate a range of improvements to credentialing and authentication technologies. As an initial matter, a digital seal system that allows for dynamic validations and invalidations according to one or more embodiments greatly simplifies current credentialing approaches which, for example, typically require new seals to be issued following an invalidation. Moreover, the digital seal itself is configured according to one or more embodiments as a multi-level signature rather than a single-level signature common to traditional digital certificates.

(22) In particular, the digital seal directs a user, upon selection of the seal in the digital space (or by scanning a QR code incorporated within the seal in the physical space), to a validation portal where the user can easily see the current validation status of the seal (i.e., whether the seal remains valid or whether the seal has been invalidated post-issuance) and if valid, also provides an option to check the single-level signature of the seal. A single-level signature includes, for example, a Public Key Infrastructure (PM) signature stored on a blockchain. If invalid, verification of the single-level signature is blocked. The validation portal also includes supplementary information, such as the validation history of the seal, the reason(s) for invalidations, etc., as well as a link to multi-level signature data. As used herein, “multi-level signature” data refers to static variables and dynamic variables digitally attached to the seal. Static variables and dynamic variables are discussed in greater detail herein, but generally refer to multi-level digital signature information that cannot be updated after digitally signing/sealing (static) and multi-level digital signature information that can be updated after digitally signing/sealing (dynamic).

(23) FIG. 1 illustrates a document **100** having a dynamically validated, multi-level signature seal **102** according to one or more embodiments. As shown in FIG. 1, the seal **102** can include various structural elements, such as, for example, an attestation **104**, seal details **106**, and machine-readable code **108**. Although the document **100** and seal **102** are shown in a particular configuration for ease

of discussion and illustration, it should be understood that the actual content of the document **100** and the seal **102** can vary depending on the needs of the respective application.

(24) The attestation **104** can include a certification, stamp, and/or seal from a bearer of the seal **102** for the document **100** and can be signed or unsigned by the bearer. Like the wax seal from antiquity, the attestation **104** serves as an easily observable accreditation from the bearer for the contents of the document **100**. In some embodiments, the attestation **104** includes a license number or other identifier for the bearer of the seal **102**.

(25) The seal details **106** can include information related to the attestation **104** and/or the seal **102**. In some embodiments, the seal details **106** can provide the name of the bearer of the seal **102**, a timestamp indicating the time that the attestation **104** was made, metadata for the seal **102** itself such as, for example, the name of the company responsible for maintaining the seal **102** multi-level signature data, and a link **110**. In some embodiments, selecting the link **110** directs the user to a validation portal (e.g., validation portal **200** in FIGS. 2A and 2B) where the user can easily see the current validation status of the seal **102**. In some embodiments, the seal **102** is configured to directly serve as the link **110**. In other words, selecting any area within the boundaries of the seal **102** can direct the user to the validation portal. Validation portals are discussed in greater detail with respect to FIGS. 2A and 2B. FIG. 2A depicts a validation portal **200** when the seal **102** is valid, and FIG. 2B depicts the validation portal **200** when the seal **102** is invalid.

(26) The machine-readable code **108** can include, for example, a barcode or a quick response (QR) code (as shown). In some embodiments, the machine-readable code **108** includes the link **110**. In some embodiments, scanning the machine-readable code **108** directs the user to the validation portal in a similar manner as selecting the link **110**. In this manner, the seal **102** can be easily validated by a user even if the document **100** (and the seal **102**) is printed out or is otherwise made available in physical form. Although depicted as a QR code for ease of illustration, in some embodiments of the invention, the machine-readable code **108** does not include a barcode, QR code, etc., but is rather code embedded within the seal **102** itself. In this manner, any portion of the seal **102** (e.g., the attestation **104**, the seal details **106**, whitespace within the seal **102**, and/or any other region of the seal **102**) can be clicked to direct the user to the validation portal in a similar manner as selecting the link **110**.

(27) FIGS. 2A and 2B illustrate a validation portal **200** reached by clicking or selecting the link **110** (or seal **102** itself) and/or scanning the machine-readable code **108** shown in FIG. 1. Although the validation portal **200** is shown in a particular configuration and as having specific elements for ease of discussion and illustration, it should be understood that the actual content of the validation portal **200** can vary depending on the needs of the respective application. For example, the validation portal **200** can include more, or less elements than shown in FIGS. 2A and 2B.

(28) FIG. 2A depicts the validation portal **200** when the seal (e.g., seal **102**) is valid. In some embodiments, the validation portal **200** is a website or webpage that the user is automatically directed to (via, e.g., embedded URLs in the link **110**) upon selection of the link **110** and/or via scanning the machine-readable code **108**. In some embodiments, the validation portal **200** is configured as a pop-up window in a user interface used to interact with the document **100**. The pop-up window can be automatically triggered upon selection of the link **110** (via, e.g., code embedded in the seal **102** and/or link **110**).

(29) As shown in FIG. 2A, the validation portal **200** includes a validation status window **202** (here, "SealPact Validation Status"). In some embodiments, the validation status window **202** includes an artifact **204** graphically indicating the validation status of the respective seal (e.g., seal **102**). In some embodiments, the artifact **204** is configured as a check mark (as shown), thumbs up, smile emoji, or other graphical object that is easily recognizable as a positive indication that the respective seal is a valid seal.

(30) As further shown in FIG. 2A, the validation status window **202** can include an identifier **206** for the seal **102**. The identifier **206** can include a uniquely identifiable digital key, name, string, or

sequence of symbols or alphanumeric characters for the seal **102**. In some embodiments, the identifier **206** is a cryptographically protected (via, e.g., hashing, public and private keys, etc.) identifier.

(31) In some embodiments, the validation status window **202** includes a single-level digital signature **208**. The single-level digital signature **208** includes a first level of authentication information for the seal **102**. In some embodiments, the single-level digital signature **208** includes a statement or status indicating that the identifier **206** (e.g., digital signature keys) is unique to the licensee indicated on the seal **102**. Alternatively, if the identifier **206** is not unique to the licensee, the single-level digital signature **208** can include a statement or status indicating such. In some embodiments, the single-level digital signature **208** includes a statement or status indicating that the document **100** has not been (or has been) altered subsequent to the application of the seal **102**. For example, the seal **102** can be configured such that any editing of the document **100** after placement of the seal **102** triggers the seal **102** to automatically update the respective message in the single-level digital signature **208** to warn users that the document **100** has been altered.

(32) In some embodiments, the validation status window **202** includes multi-level validation information **210** corresponding to multi-level digital signature data for the seal **102**. For example, multi-level validation information **210** can include a statement or status indicating that the private key used for signing electronic documents with the seal **102** is under the licensee's direct and exclusive control and the seal **102** has not been (or has been) manually invalidated by the licensee. The content of the multi-level digital signature data is discussed in greater detail below.

(33) In some embodiments, the validation status window **202** includes a single-level signature validation link **212** when the seal **102** is valid (i.e., as shown in FIG. 2A). In some embodiments, selecting the single-level signature validation link **212** directs the user to a resource (via popup, website or webpage redirects, etc.) that allows the user to manually verify the single-level digital signature **208** (e.g., that the document **100** hasn't been tampered with) using, for example, a remotely accessed resource secured via a Secure Hash Algorithm (SHA). In some embodiments, selecting the single-level signature validation link **212** directs the user to a verification pop-up window (not separately shown) configured such that the user can choose a file or drag a file to the verification pop-up window to have the respective file validated (e.g., via SHA).

(34) In some embodiments, the validation status window **202** includes a multi-level signature validation link **214** (here, "View SealPact Details"). In some embodiments, selecting the multi-level signature validation link **214** directs the user to a multi-level digital signature portal (e.g., the multi-level digital signature validation portal **300**) via a popup, website or webpage redirect, etc., that allows the user to manually verify multi-level digital signature data for the document **100** and/or the seal **102**. For example, selecting the multi-level signature validation link **214** can direct the user to one or more of FIGS. 3, 4, 5, and 6, the contents of which are discussed in greater detail below.

(35) In some embodiments, the validation status window **202** includes a live link **216** to the underlying document itself (e.g., document **100**) when the seal **102** is valid (i.e., as shown in FIG. 2A).

(36) In some embodiments, interacting with the validation portal **200** (e.g., selecting the single-level signature validation link **212**, the multi-level signature validation link **214**, the live link **216**, clicking anywhere within the validation portal **200**, etc.) directs the user (via, e.g., a pop-up or browser redirect) to a subscription window (not separately shown). In some embodiments, the subscription window allows the user to register to receive updates on edits to the respective document/project (e.g., document **100**). In some embodiments, the user can register within the subscription window by providing the user's email address (or other contact details, as desired). The subscription window can be served to users regardless of the current validation status of the seal **102**.

(37) FIG. 2B depicts the validation portal **200** when the seal (e.g., seal **102**) is invalid. The

validation portal **200** can include a validation status window **202**, an artifact **204**, an identifier **206**, a single-level digital signature **208**, a multi-level validation information **210**, and a multi-level signature validation link **214**, configured in a same or similar manner as discussed with respect to FIG. 2A. Notably, in some embodiments, the validation status window **202** does not include a single-level signature validation link **212** when the seal **102** is invalid (i.e., as shown in FIG. 2B). Similarly, in some embodiments, the validation status window **202** does not include a live link **216** to the underlying document itself (e.g., document **100**) when the seal **102** is invalid (i.e., as shown in FIG. 2B).

(38) As shown in FIG. 2B, the artifact **204** graphically indicates that the respective seal (e.g., seal **102**) is not valid. In some embodiments, the artifact **204** is configured as an “X” mark (as shown), thumbs down, frown emoji, or other graphical object that is easily recognizable as an indication that the respective seal is no longer valid. In some embodiments, the validation status window **202** is configured to visually highlight one or more reasons for the invalid status. For example, the text associated with the single-level digital signature **208** and/or multi-level validation information **210** can be highlighted, changed in color, bolded, underlined, or otherwise altered to direct a user's attention to the reason(s) for the invalid status. As shown in FIG. 2B, the seal **102** is invalid because it was manually invalidated by the licensee. This example is merely explanatory of course, and it should be understood that seals can be invalidated for a variety of reasons (e.g., digital keys are compromised, document has been altered, revised, etc., licensee does not have direct and/or exclusive control of the private key, licensee has revoked/suspended license, or for any other reason). In some embodiments, the validation status window **202** includes supplementary information **218** attached to the reason(s) for invalidation, such as, for example, a statement that the licensee manually invalidated the seal **102** due to an update or revision change in the underlying deliverable (e.g., document **100**).

(39) FIG. 3 illustrates a first page **302** (here, “1-Basic”) of a multi-level digital signature validation portal **300** reached by clicking or selecting the multi-level signature validation link **214** shown in FIGS. 2A and 2B. Although the validation portal **300** is shown in a particular configuration and as having specific elements for ease of discussion and illustration, it should be understood that the actual content and arrangement of elements of the validation portal **300** can vary depending on the needs of the respective application. For example, various elements of the validation portal **300** shown in FIG. 3 can instead be located within the other pages (views) depicted in FIGS. 4, 5, and 6 (and vice versa). Moreover, the actual number and ordering of the various pages or views of the multi-level digital signature validation portal **300** can be arbitrarily changed or customized as desired for a particular application (e.g., which page is designated the “first” page or landing upon selecting the multi-level signature validation link **214** can be changed as desired, how elements are allocated between the various pages, how many pages, etc.).

(40) As shown in FIG. 3, the first page **302** includes basic information **304** associated with the seal **102**. As used herein, “basic” information includes information that is already available in the source document (e.g., document **100**). For example, the basic information **304** can include the name of the licensee, the license number, the firm name associated with the licensee, a project name, a project number, a project address, a timestamp or a date indicating the time that the seal **102** was made, and/or any other additional information **306** already available in the source document. In some embodiments, the “basic” information is static information. As used herein, “static information” refers to information that cannot be changed by the signer or other users after the seal **102** is signed (i.e., after attestation). Advantageously, the basic information **304** provided in the first page **302** serves as a verified source of data against which a user can manually compare to information found in the document **100**. This is useful, for example, when the document **100** has been printed or photocopied. In this manner, the user can ensure that no one has altered the information found in the document **100**.

(41) In some embodiments, the first page **302** includes one or more navigation link(s), such as the

link **308** (here, “View Document”), link **310** (“2-Advanced”), link **312** (“3-Quality Control”), and link **314** (“4-Contractual Info”). In some embodiments, selecting the link **308** directs the user to the underlying document (e.g., document **100** as shown in FIG. 1). In some embodiments, selecting the links **310**, **312**, and **314** directs the user to the associated page of the multi-level digital signature validation portal **300** (e.g., FIGS. 4, 5, and 6, receptively).

(42) FIG. 4 illustrates a second page **402** (here, “2-Advanced”) of the multi-level digital signature validation portal **300**. As shown in FIG. 4, the second page **402** includes advanced information **404** associated with the seal **102**. As used herein, “advanced” information includes information that is not already available in the source document (e.g., document **100**). In some embodiments, the “advanced” information is static information. As used herein, “static information” refers to information that cannot be changed by the signer or other users after the seal **102** is signed (i.e., after attestation).

(43) In some embodiments, the advanced information **404** includes a client name, client email, client type, owner type, nature of work, discipline, concentration, project role, scope inclusions, scope exclusions, identifiers for retained sub-consultants, sub-consultant disciplines, sub-consultant concentrations, and/or any other additional information **406** that is not already available in the source document. Advantageously, the advanced information **404** serves as a verified supplemental source of data for the user. This is useful, for example, when the user is interested in the scope of liability (e.g., are there scope inclusion or exclusions?) associated with the seal **102** or in other supplemental information that is not typically included within the four corners of the document **100** (i.e., information that is not found within the physical copy of the document **100**).

(44) In some embodiments, the second page **402** includes one or more navigation link(s), such as the link **408** (here, “View Document”), link **410** (“1-Basic”), link **412** (“3-Quality Control”), and link **414** (“4-Contractual Info”). In some embodiments, selecting the link **408** directs the user to the underlying document (e.g., document **100** as shown in FIG. 1). In some embodiments, selecting the links **410**, **412**, and **414** directs the user to the associated page of the multi-level digital signature validation portal **300** (e.g., FIGS. 3, 5, and 6, receptively).

(45) FIG. 5 illustrates a third page **502** (here, “4-Contractual Info”) of the multi-level digital signature validation portal **300**. As shown in FIG. 5, the third page **502** includes contractual information **504** associated with the seal **102**. As used herein, “contractual” information includes information associated with the one or more contracts associated with the document **100** that is not already available in the source document (e.g., document **100**). In some embodiments, the “contractual” information is static information. As used herein, “static information” refers to information that cannot be changed by the signer or other users after the seal **102** is signed (i.e., after attestation).

(46) In some embodiments, the contractual information **504** includes a fee amount, a contract type, an uploaded contract link (e.g., a URL link to the respective contract), an identifier for the contract drafter, limits of liability, identifiers indicating whether there are waivers of consequential damages, and/or any other additional information **506** that is not already available in the source document. Advantageously, the contractual information **504** serves as a verified supplemental source of contractual data for the user, in a similar manner as the additional information **406** provides additional details as discussed with respect to FIG. 4.

(47) In some embodiments, the third page **502** includes one or more navigation link(s), such as the link **508** (here, “View Document”), link **510** (“1-Basic”), link **512** (“2-Advanced”), and link **514** (“3-Quality Control”). In some embodiments, selecting the link **508** directs the user to the underlying document (e.g., document **100** as shown in FIG. 1). In some embodiments, selecting the links **510**, **512**, and **514** directs the user to the associated page of the multi-level digital signature validation portal **300** (e.g., FIGS. 3, 4, and 6, receptively).

(48) FIG. 6 illustrates a fourth page **602** (here, “3-Quality Control”) of the multi-level digital signature validation portal **300**. As shown in FIG. 6, the fourth used herein, “quality control”

information includes dynamic information associated with the seal **102** that is not already available in the source document (e.g., document **100**). As used herein, “dynamic information” refers to information that can be changed or otherwise updated by the signer (or other users) after the seal **102** is signed (i.e., after attestation). For example, dynamic information can be changed or updated in a means to track construction progress, proficiency, and adequacy of the project for the subject (original) sealed artifact.

(49) In some embodiments, the quality control information **604** includes an indicator for whether quality control is needed for the respective seal (e.g., seal **102**), an indicator for whether independent review is required (as shown, “No”; however, this field can also indicate that independent review is required, e.g., “Yes”, and if so, whether it is completed or not, e.g., “Yes—Complete/Not Complete”), an indicator for whether shop drawings are required (for, e.g., stamped delegated and/or component designs) and if so, an indicator for which (here, Concrete: Mix Design—without Air, Concrete: Cast-in-Place Reinforcing, Cold-Formed Steel Framing: Metal Stud Framing/Details, Architectural: Windows, Architectural: Finishes, etc.), an indicator for whether a construction administration (CA) has been assigned to the respective matter, and if so, an indicator for a number of site visits made by the CA (here, “3”), an indicator for special instructions (SA), a link to any such SI forms, and/or any other additional information **606** that is not already available in the source document. Advantageously, the quality control information **604** serves as a verified source of dynamic information for the user.

(50) In some embodiments, the fourth page **602** includes one or more navigation link(s), such as the link **608** (here, “View Document”), link **610** (“1-Basic”), link **612** (“2-Advanced”), and link **614** (“4-Contractual Info”). In some embodiments, selecting the link **608** directs the user to the underlying document (e.g., document **100** as shown in FIG. 1). In some embodiments, selecting the links **610**, **612**, and **614** directs the user to the associated page of the multi-level digital signature validation portal **300** (e.g., FIGS. 3, 4, and 5, receptively).

(51) In contrast to the static information discussed previously with respect to FIGS. 3, 4, and 5, the dynamic information of the quality control information **604** is configured to change automatically or manually when necessary. In some embodiments, dynamic fields are linked to one or more external data sources. In some embodiments, dynamic fields are configured to continuously or periodically (at any interval set by the user) poll their respective external data source and can update automatically when the underlying information changes. For example, if the CA makes another site visit, the “Site Visits” field can automatically update to “4” by pulling site visit data from a linked external source. In another example of the dynamic nature of the quality control information **604**, the link to the SI form shown in the quality control information **604** (“Attach (SI) Form—linked SI Form”) can be configured as a dynamic hyperlink (e.g., URL) to the respective source for the SI form. In some embodiments, the quality control information **604** can be configured to dynamically updated the hyperlink when and if the source destination for the SI form changes.

(52) FIG. 7 illustrates a user interface **700** for managing dynamically validated, multi-level signature seals (e.g., seal **102**) and multi-level digital signature validation portals (e.g., multi-level digital signature validation portal **300**) according to one or more embodiments. As shown in FIG. 7, the user interface **700** (also referred to as a website or web portal) includes a navigation bar **702** and a dashboard **704** (here, “My Dashboard”).

(53) In some embodiments, the navigation bar **702** includes one or more selectable fields **706** (here, “My Dashboard”, “My Seals”, “My Documents”, “My Account”, “Subscription”). In some embodiments, clicking or otherwise selecting a field of the one or more selectable fields **706** redirects the dashboard **704** to content associated with the respective field. For example, in FIG. 7 “My Dashboard” has been selected and the dashboard **704** now displays information associated with “My Dashboard”, such as a seal widget **708** including a list of managed seals for the respective user of the user interface **700**. The information can be displayed via widgets, popups,

and/or windows within the user interface **700**.

(54) In some embodiments, “My Dashboard” serves as a landing page for the user interface **700** that offers a high-level overview of the various seals and associated documents for the respective user. The exact type and scope of the data presented in “My Dashboard” is not meant to be particularly limited, but could further include, for example, a signatures widget **710**, a stamps widget **712**, and a customizable widget **714**.

(55) In some embodiments, the signatures widget **710** includes a list of managed signatures for the respective user of the user interface **700**. In some embodiments, the stamps widget **710** includes a list of managed stamps for the respective user of the user interface. In some embodiments, the customizable widget **710** can be customized by a user of the user interface **700** to include any available information (e.g., a count of managed seals/stamps/signatures for the user, a list of associated documents, contract or liability information, etc.). In some embodiments, “My Dashboard” further includes a navigation widget **716** configured for uploading and searching documents (e.g., document **100** having seal **102**).

(56) FIG. **8** illustrates an exemplary selectable field **802** of the user interface **700** according to one or more embodiments. As shown in FIG. **8**, the selectable field **802** (here, “My Documents”) has been selected and the dashboard **704** now displays information associated with “My Documents”, such as a document widget **804**. While not meant to be particularly limited, the document widget **804** can include various informational fields for one or more documents managed using the user interface **700**. For example, the document widget **804** includes fields for “Signer Name”, “Type”, “Project Number”, “Project Address”, “Project Name”, and “Date Uploaded”. It should be understood that other informational fields are possible. The various informational fields of the document widget **804** can be selected to sort (alphanumerically, ascending or descending, etc.) the associated documents, seals, and stamps managed using the user interface **700**. Other widgets are possible. For example, managers can use a signer widget **806** to quickly change (via, e.g., a pulldown menu) between various signers, causing the user interface **700** to display the seals, stamps, and documents associated with the selected signer.

(57) In some embodiments, selecting any line within the document widget **804** causes the dashboard **704** to navigate to the associated document. For example, selecting the line **808** causes the dashboard **704** to navigate to the document **100** (shown in FIG. **1**). An example document navigation is shown in FIG. **9A**.

(58) FIGS. **9A**, **9B**, and **9C** illustrate the dashboard **704** after selecting a document **902** from the document widget **804** according to one or more embodiments. As shown in FIG. **9A**, the dashboard **704** shows a page **904** (here, an example title page) of the document **902** having thereon a seal **906**. As further shown in FIG. **9A**, the dashboard **704** now includes a navigation widget **908**.

(59) The navigation widget **908** can include various tools for navigating the document **902**. As shown, the navigation widget **908** includes a download button that upon selection, allows the user to download a local copy of the document **902**, a page count indicator, and various page navigation buttons (e.g., page up/down buttons, move, zoom and rotate buttons, etc.).

(60) The navigation widget **908** further includes a validation widget **910** and a seal status widget **912**. The validation widget **910** can be selected by the user to manually validate seals and to manually invalidate previously valid seals. The seal status widget **912** displays the current validation status for the respective document seal (here, the validation status for the seal **906** on the document **902**).

(61) FIG. **9A** illustrates an example scenario where the seal **906** is currently valid (sealed, stamped, etc.). Accordingly, the validation widget **910** displays an option to “Invalidate” the seal **906** and the seal status widget **912** indicates “SealPacted” (or alternatively, any other language to positively affirm that the seal is valid, such as, for example, “Valid”, “Sealed”, “Signed”, “Stamped”, etc.).

(62) In some embodiments, the validation widget **910** is only shown to users having authority to change the seal status for the respective document. For example, the validation widget **910** can be

shown only when the currently logged in user within the user interface **700** is the signer for the respective document. Permissions can be managed using any suitable technique, such as, for example via permissions tied to individual accounts assigned to each signer or user of the user interface **700**. These accounts can be secured via logins and passwords, 2-factor authentication, etc. In any case, for users that do not have signer permissions, the dashboard **704** can be configured to only display the seal status widget **912**.

(63) FIG. **9B** illustrates an example scenario where the seal **906** shown in FIG. **9A** needs to be invalidated. As shown in FIG. **9B**, the validation widget **910** can include a pulldown menu with various reasons or explanations to “Invalidate” the seal **906**. While the options shown in FIG. **9B** are illustrative only and are not meant to be particularly limited, such options can include, for example, “Updates/Revisions to Deliverable”, “Updates/Revisions to SealPact information”, “Design Professional and Client have Terminated Relationship”, “Other”, etc.

(64) FIG. **9C** illustrates an example scenario where the seal **906** shown in FIG. **9A** has been invalidated. As shown in FIG. **9C**, the validation widget **910** now displays “Re-SealPact” (or alternatively, any other language to positively affirm that the seal can be re-validated, such as, for example, “Validate”, “Seal”, “Re-seal”, “Sign”, “Re-sign”, “Stamp”, “Revalidate”, etc.). Moreover, the seal status widget **912** now indicates “Invalidated” (or alternatively, any other language to positively affirm that the seal is now invalid, such as, for example, “Invalid”, “Not Sealed”, “Not Signed”, “Stamp Revoked”, etc.).

(65) FIG. **10** illustrates aspects of an embodiment of a computer system **1000** that can perform various aspects of embodiments described herein. The computer system **1000** includes at least one processing device **1002**, which generally includes one or more processors for performing aspects of methods described herein.

(66) Components of the computer system **1000** include the processing device **1002** (such as one or more processors or processing units), a memory **1004**, and a bus **1006** that couples various system components including the system memory **1004** to the processing device **1002**. The system memory **1004** may include a variety of computer system readable media. Such media can be any available media that is accessible by the processing device **1002**, and includes both volatile and non-volatile media, and removable and non-removable media.

(67) For example, the system memory **1004** includes a non-volatile memory **1008** such as a hard drive, and may also include a volatile memory **1010**, such as random access memory (RAM) and/or cache memory. The computer system **1000** can further include other removable/non-removable, volatile/non-volatile computer system storage media.

(68) The system memory **1004** can include at least one program product having a set (e.g., at least one) of program modules that are configured to carry out functions of the embodiments described herein. For example, the system memory **1004** stores various program modules that generally carry out the functions and/or methodologies of embodiments described herein. A module or modules **1012**, **1014** may be included to perform functions related to the seal **102**, validation portal **200**, user interface **700**, etc., such as, for example, validating, invalidating, or re-validating the seal **102**, adding, removing, or amending any of the static or dynamic multi-level signature data, etc. The system **1000** is not so limited, as other modules may be included depending on the functionality required for a respective application. As used herein, the term “module” refers to processing circuitry that may include an application specific integrated circuit (ASIC), an electronic circuit, a processor (shared, dedicated, or group) and memory that executes one or more software or firmware programs, a combinational logic circuit, and/or other suitable components that provide the described functionality.

(69) The processing device **1002** can also be configured to communicate with one or more external devices **1016** such as, for example, a keyboard, a pointing device, and/or any devices (e.g., a smartphone, a laptop, a stylus, etc.) that enable the processing device **1002** to communicate with one or more other computing devices. Communication with various devices can occur via

Input/Output (I/O) interfaces **1018** and **1020**.

(70) The processing device **1002** may also communicate with one or more networks **1022** such as a local area network (LAN), a general wide area network (WAN), a bus network and/or a public network (e.g., the Internet) via a network adapter **1024**. In some embodiments, the network adapter **1024** is or includes an optical network adaptor for communication over an optical network or near-field communication (NFC). It should be understood that although not shown, other hardware and/or software components may be used in conjunction with the computer system **1000**. Examples include, but are not limited to: microcode, device drivers, redundant processing units, external disk drive arrays, RAID systems, and data archival storage systems, etc.

(71) Referring now to FIG. **11**, a flowchart **1100** for leveraging a digital signature system having dynamically validated, multi-level signature seals is generally shown according to an embodiment. The flowchart **1100** is described in reference to FIGS. **1-10** and may include additional steps not depicted in FIG. **11**. Although depicted in a particular order, the blocks depicted in FIG. **11** can be rearranged, subdivided, and/or combined.

(72) At block **1102**, a document having an attestation made by a conveyor is received. The attestation can include, for example, a seal, stamp, or certification made by an authorized person(s) (e.g., a notary, a designated signor, etc.). In some embodiments, the document and attestation are digital facsimiles of physical documents and attestations (e.g., a stamp, seal, etc.).

(73) At block **1104**, machine-readable code is attached to at least one of the document and the attestation. The machine-readable code can be configured to direct a user, upon scanning or selecting the machine-readable code, to multi-level digital signature data for the attestation. In some embodiments, the multi-level digital signature data includes an artifact configured to graphically indicate the validation status of the respective attestation (e.g., seal). In some embodiments, the multi-level digital signature data includes static information that is available in the document, static information that is not available in the document, and dynamic information that is not available in the document (sometimes collectively referred to as multi-level validation data).

(74) The method can further include scanning, by a user device of the user, the machine-readable code and directing a user interface of the user device to a resource (e.g., a website, portal, or popup having the validation information). The resource can include the artifact and the multi-level signature data. The user can be directed to the resource/data via directing the device to a website, portal, or dashboard having therein the multi-level digital signature data.

(75) The method can further include receiving credentials showing that the user is the conveyor. For example, the user can login to a credentialing system using a known account name and password, via a private key, etc.

(76) The method can further include receiving, from the user (after authentication), a manual invalidation of the attestation from the user interface of the user device. In some embodiments, the user manually invalidates the seal/attestation via a user interface configured to display the multi-level digital signature data. In some embodiments, the user can select a reason for invalidating the attestation from a dropdown in the user interface. In some embodiments, the user can manually re-validate the attestation via the user interface. In some embodiments, the user can select a reason for re-validating the attestation from a dropdown in the user interface.

(77) While the above disclosure has been described with reference to exemplary embodiments, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted for elements thereof without departing from its scope. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the disclosure without departing from the essential scope thereof. Therefore, it is intended that the present disclosure not be limited to the particular embodiments disclosed, but will include all embodiments falling within the scope thereof

Claims

1. A computer system including one or more processors, a memory, and a bus that couples the one or more processors including the memory to each other, the memory stores one or programs to define a A digital signature system comprising: a multi-level signature seal comprising: an attestation for a document made by a conveyor; and machine-readable code configured to direct a user, upon scanning or selecting the machine-readable code, to a validation portal for the attestation; the validation portal comprising: an artifact graphically indicating a current validation status of the attestation; and a validation link configured to direct the user, upon selection, to multi-level signature data comprising static information that is available in the document, static information that is not available in the document, and dynamic information that is not available in the document; and a user dashboard configured to allow the conveyor to manually validate and invalidate the attestation.
2. The computer system of claim 1, wherein the attestation comprises at least one of a seal, a stamp, and a certification made on authority of the conveyor.
3. The computer system of claim 1, wherein the validation portal further comprises a unique identifier for the multi-level signature seal.
4. The computer system of claim 1, wherein the validation portal further comprises a single-level digital signature and a link configured to direct the user, upon selection, to a cryptographically protected resource for manually verifying the single-level digital signature.
5. The computer system of claim 4, wherein the validation portal further comprises at least one multi-level validation statement.
6. The computer system of claim 5, wherein the at least one multi-level validation statement comprises an affirmation that a private key used for signing documents with the multi-level signature seal is under the conveyor's direct and exclusive control.
7. The computer system of claim 6, wherein the at least one multi-level validation statement comprises an affirmation that the multi-level signature seal has not been manually invalidated by the conveyor.
8. The computer system of claim 6, wherein the at least one multi-level validation statement comprises an affirmation that the multi-level signature seal has been manually invalidated by the conveyor.
9. The computer system of claim 8, wherein the at least one multi-level validation statement further comprises one or more reasons for manually invalidating the multi-level signature seal.
10. The computer system of claim 1, wherein the machine-readable code comprises a barcode or quick response (QR) code.
11. A computer system including one or more processors, a memory, and a bus that couples the one or more processors including the memory to each other, the memory stores one or programs to define a A multi-level signature seal comprising: an attestation for a document made by a conveyor; and machine-readable code configured to direct a user, upon scanning or selecting the machine-readable code, to: an artifact graphically indicating a current validation status of the attestation; and multi-level signature data comprising static information that is available in the document, static information that is not available in the document, and dynamic information that is not available in the document.
12. The computer system of claim 11, wherein the attestation comprises at least one of a seal, a stamp, and a certification made on authority of the conveyor.
13. The computer system of claim 11, wherein the machine-readable code comprises a barcode or quick response (QR) code.
14. The computer system of claim 11, wherein the machine-readable code comprises a selectable link.

15. A method comprising: receiving a document comprising an attestation made by a conveyor; and attaching machine-readable code to at least one of the document and the attestation, the machine-readable code configured to direct a user, upon scanning the machine-readable code, to: an artifact graphically indicating a current validation status of the attestation; and multi-level signature data comprising static information that is available in the document, static information that is not available in the document, and dynamic information that is not available in the document.
16. The method of claim 15, wherein the attestation comprises at least one of a seal, a stamp, and a certification made on authority of the conveyor.
17. The method of claim 15, wherein the machine-readable code comprises a barcode or quick response (QR) code.
18. The method of claim 15, further comprising: scanning, by a user device of the user, the machine-readable code; and directing a user interface of the user device to a resource comprising the artifact and the multi-level signature data.
19. The method of claim 18, further comprising receiving credentials showing that the user is the conveyor.
20. The method of claim 19, further comprising receiving, from the user, a manual invalidation of the attestation from the user interface of the user device.
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